

Exploring the Evolving Landscape of API Security Challenges in the Healthcare Industry: A Comprehensive Review

1st Akram Al-Rumaim
Goa Business School(GBS)
Goa University
Goa, India
akramalrumaim@gmail.com

2nd Jyoti D. Pawar
Goa Business School(GBS)
Goa University
Goa, India
jdp@unigoa.ac.in

Abstract—This academic investigation offers a thorough exploration of API security challenges in the Healthcare sector. Through a meticulous review of existing literature, it identifies vulnerabilities and threats, including complex authentication issues, data leakage risks, injection attacks, and weaknesses in access control. The study also points out deficiencies in current security practices, such as improper documentation and tracking protocols. This research provides critical observation for improving healthcare API security and protecting patient data. By lobbying for reliable security practices, Key Players can successfully address these risks. Finally, this paper sheds light on the API security concerns in healthcare, allowing organizations to strengthen healthcare APIs and preserve sensitive patient information, with success validated by quantitative findings.

Index Terms—Cybersecurity, API Security, Cyber Infrastructure Protection, Cyber threats, Healthcare API

I. INTRODUCTION

In the rapidly advancing landscape of healthcare digitization, the integration of Application Programming Interfaces (APIs) has emerged as a transformative force, enabling seamless data exchange, interoperability, and improved patient care. APIs facilitate efficient communication and information sharing between diverse healthcare systems and stakeholders, fostering innovation and enhancing operational efficiency within the Healthcare industry. However, this unprecedented level of connectivity and data accessibility also brings forth a host of security challenges that demand rigorous attention.

Ensuring the security and privacy of patient data has become a paramount concern for healthcare organizations. The growing reliance on APIs exposes the Healthcare industry to a range of potential vulnerabilities and threats, including unauthorized access, data breaches, and injection attacks. As a result, safeguarding sensitive patient information and maintaining the integrity of healthcare systems have become critical imperatives in the modern digital era.

This research aims to explore the intricacies and complexities of API security within the Healthcare industry. By conducting a comprehensive review, the study seeks to identify and assess various dimensions of API vulnerabilities and threats encountered by healthcare organizations. Furthermore,

it will evaluate the effectiveness of existing API security measures and practices adopted by healthcare systems.

By shedding light on the challenges and potential risks surrounding API security, this research endeavors to equip healthcare professionals, policymakers, and technologists with invaluable insights. Implementing proactive and robust security measures will be crucial in safeguarding patient data and preserving the trust of individuals seeking quality healthcare services. In a world increasingly dependent on digital technologies, this research contributes to fortifying the Healthcare industry against the ever-evolving landscape of cyber threats, thus ensuring the seamless and secure functioning of healthcare API ecosystems.

The subsequent segments of this manuscript are organized as follows:

In Section 2, a comprehensive exposition of commonly utilized Application Programming Interfaces (APIs) is provided, accompanied by an exploration of potential vulnerabilities and associated security risks. Section 3 presents an exhaustive review of existing literature concerning API Security within the Healthcare context. This includes a detailed overview of research findings and an exploration of any identified limitations. The conclusions drawn from this study are outlined in Section 4, emphasizing the significant insights derived from the research. Finally, Section 5 elucidates potential avenues for future research in this field, thereby highlighting areas that warrant further investigation.

II. HEALTHCARE API(APPLICATIONS AND POTENTIAL RISKS)

The uses of Healthcare APIs are manifold and diverse, allowing healthcare organizations to streamline processes, access patient records, and facilitate data-driven decision-making. These APIs enable secure authentication and authorization processes, ensuring that only authorized personnel can access patient data. They also facilitate the exchange of Electronic Health Records (EHRs) between healthcare providers, enhancing patient care coordination and reducing duplication of efforts.

TABLE I
API TYPES (THEIR USES AND POTENTIAL THREATS)

API Type	Uses	Example	Potential Threat
FHIR(Fast Healthcare Interoperability Resources) API	-Facilitates electronic data exchange	-SMART on FHIR: Enables apps to access patient data from EHRs	-Data Breaches
EHR (Electronic Health Record) APIs	-Enables access to patient health information	-Cerner EHR API: Access patient records and medical history	-SQL Injection and Code Injection
DICOM (Digital Imaging and Communications in Medicine) API	-Facilitates medical imaging data sharing	-PACS Integration: Allows viewing and sharing medical images	-Man-in-the-Middle Attacks
Patient Portal APIs	-Empowers patients to access health records	-MyChart API: Patient portal for accessing health records	-Denial of Service (DoS) Attacks
Prescription and Medication APIs	-Enables electronic prescription transmission	-Surescripts API: Facilitates electronic prescribing	-Insider Threats
Laboratory APIs	-Integrates lab test results into health records and not only limited that	-LabCorp API: Provides lab results integration	-Lack of Proper Authentication and Authorization

According to OWASP (Open Web Application Security Project), several potential security threats and attacks can target Healthcare APIs. It is crucial to be aware of these risks to implement robust security measures and protect sensitive patient data. Below are the highly potential attacks associated with the Healthcare APIs:

1. FHIR (Fast Healthcare Interoperability Resources) API:
 - Data Breaches: Unauthorized access or leakage of patient data may occur, compromising patient privacy and confidentiality.
 - Injection Attacks: Malicious actors can attempt SQL injection or code injection to exploit vulnerabilities and gain unauthorized access to the system.
2. EHR (Electronic Health Record) API:
 - SQL Injection and Code Injection: Attackers may exploit input validation vulnerabilities to execute malicious code or manipulate data.
 - Insider Threats: Internal personnel with access to API credentials can misuse privileges to access and leak patient information.
3. DICOM (Digital Imaging and Communications in Medicine) API:
 - Man-in-the-Middle Attacks: Malicious entities can intercept and manipulate data exchanged between medical imaging devices and systems.
 - Unauthorized Access: Insufficient access controls may allow unauthorized users to view sensitive medical images.
4. Patient Portal API:
 - Denial of Service (DoS) Attacks: Attackers may overload the API with excessive requests, causing disruptions and denying legitimate users access.
 - Data Leakage: Inadequate security controls could lead to unauthorized access to patient records and personal information.
5. Prescription and Medication API:
 - Insider Threats: Internal employees with access to API credentials could misuse privileges to tamper with prescription data.

- Injection Attacks: Lack of input validation may allow attackers to inject malicious code into prescription information.
6. Laboratory API:
 - Lack of Proper Authentication: Insufficient authentication mechanisms may result in unauthorized access to laboratory test results.
 - Data Manipulation: Attackers might attempt to modify lab test data, leading to misdiagnoses or incorrect treatments.

It is essential for healthcare organizations, developers, and stakeholders to address these potential attacks by implementing stringent security measures. Measures such as input validation, encryption, strong authentication, access controls, and regular security assessments can help safeguard Healthcare APIs from potential threats and ensure the integrity and privacy of patient data. Staying vigilant and following industry best practices are key to maintaining a secure and trustworthy healthcare ecosystem. Table 1 Explains healthcare API, uses, examples and their potential attack.

Fig. 1. illustrates a threat model flow diagram showing the potential security threats and vulnerabilities within a healthcare system, particularly in the context of data exchange and API usage. This threat model provides a structured framework for understanding and addressing security concerns. So it identifies key components of a healthcare system, potential threat actors, and the arrows (edges) in the diagram which represent potential threats, indicating which threat actors may target specific components.

III. LITERATURE REVIEW

Ensuring seamless compatibility continues to be a paramount issue confronting healthcare institutions in the present era [1]. Despite the strides achieved through digital modernization and the adoption of Application Programming Interfaces (APIs), patients continue to grapple with the need to access distinct patient portals for each healthcare facility they engage with. Numerous healthcare systems encounter difficulties when it comes to efficiently sharing Electronic Health Record (EHR) information, leading to fragmented

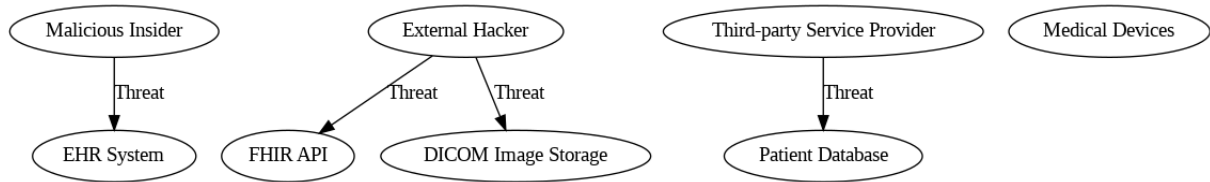


Fig. 1. Threat Model

patient information. APIs have surfaced as a viable solution for consolidating patient information, encompassing medical background and therapeutic documentation, among different healthcare platforms. Additionally, mobile applications leverage APIs to collect information from diverse medical wearable devices and seamlessly incorporate it into a patient's health dossier. However, the utilization of APIs exposes application logic and sensitive data, increasing the risk of patient data exposure to potential attackers. As the demand for tighter API security grows, managing APIs becomes crucial to safeguarding patient information and maintaining data privacy. This paper aims to provide an in-depth overview of healthcare API data integration and interoperability, while also identifying key research questions to further our understanding and knowledge base on API security in the healthcare industry. The strength lies in providing valuable insights into API data integration and its potential benefits for patients' medical records. However, it lacks a comprehensive exploration of specific security measures or solutions to address API security challenges, limiting its impact in the healthcare context.

This textbook [2] presents a comprehensive view of the US healthcare delivery system, focusing on its systemic challenges and the historical efforts to implement informatics tools to overcome these issues. With the widespread adoption of Electronic Health Record (EHR) systems, the HL7 Fast Healthcare Interoperability Resources (FHIR) API has gained rapid acceptance as the means to access data stored in these systems. The book explores the potential future state of health informatics, considering the deployment of FHIR and the increasing use of analytics to gain insights from aggregated clinical data. Notably, FHIR's adoption has led to the development of a 'universal health app platform,' fostering innovation in healthcare data utilization, similar to the app ecosystems created around smartphones. FHIR app stores have been launched by major EHR vendors, further propelling the implementation of provider-facing apps to improve EHR usability and support personalized medicine. Additionally, Medicare, the Veteran's Administration, and health insurance companies are actively exploring applications of FHIR to enhance services and communication with providers and patients. The paper's audience comprises introductory health informatics courses for both computer science and health sciences students, professionals in the health informatics community, and IT professionals seeking knowledge about the field and its implications for improved care, public/population health, and research. Despite its strengths in exploring FHIR's transformative potential, the

paper falls short in discussing the specific security challenges that accompany its widespread adoption in healthcare.

The emergence of the Internet of Things (IoT) has introduced a new paradigm in healthcare applications, [3] enhancing efficiency and services in the sector. However, this increased connectivity and communication ability of IoT devices have also brought new challenges, especially in the realm of healthcare IoT applications. Concurrently, Blockchain Technology has emerged as a potential solution in various applicable scenarios, including the healthcare industry. This chapter provides a comprehensive overview of IoT health technologies, with a specific focus on those leveraging Blockchain Technology. It explores the positive aspects of such implementations and also identifies shortcomings and future challenges in this novel research area. By surveying the latest works in this specific field, the paper aims to provide researchers with an unbiased view and valuable insights to navigate the potential of Blockchain Technology in healthcare and IoT security. The paper's strength lies in providing an unbiased perspective, aiding researchers in navigating this evolving field. However, it lacks concrete case studies or real-world applications of Blockchain Technology in healthcare, limiting its direct relevance to API security challenges.

In recent years, [4] the healthcare industry has faced serious cyber threats in the form of malware attacks targeting sensitive data and information. With the increasing adoption of virtual and cloud environments to store vast amounts of medical data, healthcare organizations are now more vulnerable to targeted attacks. Consequently, safeguarding medical data has risen to the forefront of priorities. This paper suggests a proficient strategy for detecting and categorizing malware through the application of machine learning methods. By employing dynamic features and custom sandbox analysis, the proposed scheme aims to achieve high accuracy in detecting advanced malware and attacker behavior, particularly in healthcare records. The experimental findings substantiate the superior performance and efficacy of the suggested approach in comparison to analogous methodologies. The strength of the paper lies in its technical prowess and the use of dynamic features for accurate malware detection. However, it could benefit from explicitly discussing its application to API security challenges in the healthcare industry.

The healthcare industry's increasing reliance on Internet of Things (IoT) devices [5] to collect and manage patient data has led to a rise in ransomware attacks targeting healthcare systems. To proactively protect healthcare IoT environments

from such attacks, this study proposes an early prediction scheme of ransomware behavior. By analyzing behaviors for a short period of 30 seconds and utilizing a hybrid deep learning model, the scheme successfully predicts future 90 seconds of API calls sequence associated with ransomware behavior. The experimental results demonstrate the scheme's efficacy in early detection, even with a reduced analysis time of 30 seconds, thereby reducing the likelihood of systems being infected during behavioral analysis. Its strength lies in its early detection capabilities. However, it should establish a clearer connection between ransomware behavior prediction and API security challenges in healthcare.

The increasing frequency of malware attacks on smart healthcare systems [6] necessitates robust security measures to protect sensitive data. This work proposes a novel Multi-View attention-based deep learning framework for malware detection by considering multiple features from portable executable (PE) files, including PE-Header, PE-Imports, PE-Image, and API calls. The detailed evaluation and experimental analysis demonstrate the framework's superior performance compared to other machine learning-based and non-attention-based approaches. The robustness and generalizability of the proposed method are evident in the accurate detection of malware in both Windows and Android-based smart healthcare systems. Its strength lies in addressing malware detection in healthcare systems effectively. However, it lacks a direct discussion of API security challenges specific to healthcare, limiting its relevance in the target domain.

The secure exchange of medical information [7] is crucial for improving the quality of healthcare. Blockchain technology has emerged as a potential solution to address the challenges of interoperability and security in the healthcare ecosystem. This paper focuses on exploring architectural mechanisms used to support the interoperability and security of Blockchain-based Health Management Systems (HMS). Through a systematic literature review, the paper identifies various scenarios where these mechanisms can be applied, along with their context, issues, and architectural concerns. Additionally, the paper proposes a high-level architecture and validates it through an experiment using Domain Specific Language and Model Driven Engineering for specific Smart Contracts. Its strength lies in offering valuable insights into the potential of Blockchain for healthcare interoperability and security. However, it should discuss specific API security challenges faced by healthcare organizations to enhance its impact.

APIs play a crucial role in digital communications, enabling seamless interactions between businesses and consumers. However, [8] ensuring API security remains a challenge, as vulnerabilities in API transactions can lead to cyberattacks. This research investigates the vulnerability of APIs by exploring different perspectives among customers and cyber security vendors/suppliers. The study reveals that API exploits are often undetected during their occurrence, and perspectives about security readiness differ among IT roles. The research emphasizes the importance of implementing robust API management and security programs, considering

the open and easily implementable nature of APIs in B2B and B2C communications. The strength lies in its emphasis on proactive security measures. However, it could benefit from discussing specific API security challenges unique to the healthcare industry.

This research paper presents an innovative approach [9] to bolstering API security in healthcare, particularly for Microservice Architecture Application Programming Interfaces (APIs) within the Internet of Medical Things (IoMT). The study introduces a hybrid security solution, termed SFTSDH, which seamlessly integrates Spring Framework (SF), Services for Sensitive Data (TSD), and HTTP security methods. This combination empowers healthcare organizations to implement robust identity verification, OAuth2 authorization, multifactor authentication, and access control mechanisms, ensuring compliance with the stringent General Data Protection Regulation (GDPR). The successful implementation of SFTSDH within an electronic coaching (eCoaching) prototype system showcases its effectiveness in securely collecting and managing personal health data through APIs. The paper's qualitative comparison of various security solutions, where SFTSDH emerges as promising, underscores its relevance and potential for enhancing API security in the IoMT domain. However, the paper could further explore potential challenges and scalability concerns in deploying the SFTSDH solution in larger IoMT ecosystems to provide a more comprehensive view of its applicability in diverse healthcare settings.

Continuing with a focus on securing data collection and exchange in healthcare, this study [10] implements an integrated security solution using Spring Security and Keycloak open-access platform. The solution ensures open authorization, multi-factor authentication, identity brokering, user management, and additional security measures to protect microservice APIs. The paper provides a thorough description of the implemented security solution, showcasing its efficacy in securing the eCoach prototype system from attacks. Its strength lies in the successful integration of Spring Security and Keycloak for healthcare APIs. However, it should discuss specific API security challenges faced by healthcare organizations to enhance its impact.

This survey paper [11] provides an in-depth exploration of security and privacy principles in the context of API security within the Internet of Medical Things (IoMT). It emphasizes the paramount importance of establishing robust security measures to safeguard sensitive patient information, which is frequently accessed and exchanged through APIs in healthcare settings. The introduction of a novel taxonomy based on "govern," "protect," and "detect" cyber security principles enhances the categorization of API security strategies in IoMT. The paper acknowledges the existing challenges and highlights the need for further research in the domain to fortify API security effectively. However, to bridge the gap between theoretical principles and real-world implementations, the paper could benefit from discussing practical case studies and concrete examples that illustrate API security challenges in actual healthcare environments.

Other authors dedicated their paper [12] to addressing the crucial issue of API security in the healthcare domain, particularly concerning the Internet of Medical Things (IoMT). The study proposes a scalable hybrid deep learning (DL)-driven framework, incorporating software-defined networking (SDN), to efficiently detect and combat sophisticated malwares that pose significant threats to IoMT environments. By leveraging advanced DL algorithms and SDN technology, the proposed framework exhibits superior accuracy in detecting malicious activities and efficiently mitigates potential security breaches. The evaluation of the mechanism, using publicly available datasets, validates its effectiveness in enhancing API security within IoMT. However, for broader applicability and practical implementation, further exploration of diverse IoMT environments and real-world testing scenarios is warranted.

This research primarily centers on exploring the security vulnerabilities associated with the Internet of Things (IoT), particularly within the scope of network intrusion detection systems (NIDS) integrated into IoT environments [13]. The primary objectives of the authors encompass the classification of security threats in the IoT realm, the evaluation of existing defense mechanisms, and a comprehensive examination of NIDS implementations, notably emphasizing the utilization of machine learning techniques. The study effectively categorizes threats pertinent to IoT, conducts a thorough assessment of currently available NIDS tools and datasets, and offers an extensive analysis of cutting-edge NIDS proposals tailored for IoT. Furthermore, the study underscores the efficacy of machine learning-driven NIDS in mitigating IoT-related security challenges. While the research does not directly address healthcare, its findings bear significance for the healthcare sector, which is increasingly adopting IoT devices. It indirectly underscores the critical importance of implementing robust security measures, including safeguarding Application Programming Interfaces (APIs), to fortify the protection of sensitive healthcare data. It's worth noting that this paper has a narrower focus on NIDS and machine learning techniques, possibly overlooking other dimensions of IoT security.

This research work [14], delves into the intricate security challenges presented by Android devices, with a particular focus on the pervasive issue of malware. It thoroughly examines the surge in Android malware incidents from 2010 to 2014, shedding light on the inadequacies of conventional security methodologies. Furthermore, the paper advocates for the adoption of behavior-, anomaly-, and dynamic-analysis-based techniques for malware detection, recognizing the evolving landscape of malicious threats. Notably, the study underscores the limitations inherent in traditional security approaches when confronted with Android malware, thereby stressing the imperative for a multifaceted strategy to counter advanced malware tactics effectively. Given the widespread utilization of Android devices in healthcare, catering to a multitude of applications, the paper implicitly emphasizes the critical significance of robust security measures, including the fortification of Application Programming Interfaces (APIs), within healthcare-oriented apps and devices. However, it is important to note that

the paper may offer limited technical intricacies or prescriptive remedies concerning Android security.

This paper [15] introduces "AndroSimilar," a method for generating robust signatures to detect malicious Android apps. It emphasizes the importance of signature-based solutions in countering malware variants resulting from code obfuscation and repackaging. The paper proposes a signature-based approach that is effective against code obfuscation and repackaging, common techniques used to propagate Android malware variants. Malicious Android apps can pose significant security risks to healthcare apps and APIs as well. This paper's findings highlight the importance of signature-based solutions in identifying and mitigating Android malware, which ultimately impact API security in the healthcare industry, however Signature-based methods may have limitations in detecting highly sophisticated malware.

This research [16], addresses the challenge of detecting portable executable (PE) malware by analyzing API call-grams extracted from the control flow graph of executables. It emphasizes the limitations of traditional syntactic signature-based detection methods. The paper proposes a semantic approach based on API call-grams to detect unknown malicious code, highlighting the advantages of static analysis over dynamic methods for comprehensive coverage. This paper's findings on semantic analysis for malware detection are relevant to healthcare, as it underscores the need for advanced security measures, including API security, to protect healthcare systems from PE malware threats, on the other hand, the effectiveness of semantic analysis may depend on the diversity of malware samples.

This paper evaluates the resilience of anti-malware techniques against code obfuscation in Android apps [17]. It explores transformations, including variable compression and register renaming, and assesses the effectiveness of existing anti-malware tools. The paper highlights the low resilience of anti-malware detection engines against code obfuscation, indicating the need for improved detection methods in the context of Android security. Android is widely used in healthcare environments, and this paper's findings emphasize the challenges of defending against code obfuscation techniques, indirectly highlighting the importance of strong API security in healthcare apps, but the effectiveness of anti-malware techniques can vary based on evolving obfuscation methods.

The author here, explores code obfuscation techniques in Android and their implications for security [18]. It discusses the use of obfuscation by both malware authors and legitimate developers and evaluates code de-obfuscation tools. The paper reviews code obfuscation methods, app protection techniques, and de-obfuscation methods. It emphasizes the need to investigate the efficiency of defense techniques used for code protection. As Android is prevalent in healthcare, this paper's insights into code obfuscation highlight the importance of robust API security to protect sensitive healthcare data and applications, however the challenges of code obfuscation may not have straightforward solutions.

This study presents "DroidAnalyst," [19] an automated

framework for app vetting and malware analysis on Android devices. It integrates static and dynamic analysis to improve accuracy and efficiency. The paper introduces a unified analysis model that combines static and dynamic analysis to enhance app code analysis, providing researchers and practitioners with a valuable tool for analyzing Android apps. The paper indirectly highlights the need for automated analysis techniques, which may API security assessments when it comes to healthcare sector, to ensure the security of healthcare apps running on Android devices. Possible limitation of this work could include the potential for false positives in malware detection and the need for continuous updates to adapt to the evolving Android ecosystem.

This research paper [20], addresses the challenge of detecting malicious portable executable (PE) files by analyzing API call-grams. It emphasizes the importance of behavioral analysis to detect unknown, encrypted, and zero-day malware. The paper proposes a behavior model that focuses on extracting temporal snapshots of malware and benign executables using API call-grams, providing a means to detect dynamic and encrypted malware. This paper's findings are relevant to healthcare, as they emphasize the importance of behavioral analysis in detecting malware threats that can compromise the security of healthcare APIs and systems, but behavioral analysis methods may have an increased false alarm rate.

The reviewed papers contribute valuable insights into API security challenges in the healthcare industry. Each paper offers unique approaches and solutions to tackle the security concerns surrounding API usage in healthcare settings. While some papers excel in practical implementation and real-world validation, others focus on theoretical frameworks and architectural mechanisms. Combining the strengths of these studies can pave the way for comprehensive and robust API security solutions in the healthcare sector. However, a more direct discussion of API security challenges specific to healthcare settings in some papers could further enhance their impact on the target domain. Overall, the research presented in these papers constitutes a significant step towards fortifying API security in the healthcare industry. Table 2 shows a summary of the literature reviewed stating the main focus, key-findings, and limitations.

IV. CONCLUSION

The primary objective of this research was to conduct a comprehensive examination of API security challenges within the Healthcare industry. We undertook this study to address the pressing need for a thorough understanding of these challenges and their implications for safeguarding patient data and maintaining healthcare system integrity.

Through a meticulous review of existing literature, we successfully identified and scrutinized vulnerabilities, threats, and gaps in API security. Our investigation aimed to underscore the critical importance of addressing these concerns to protect sensitive information and uphold patient privacy. Key-findings of our study revealed authentication complexities, data leakage risks, injection attacks, and access control weaknesses

as prominent challenges in healthcare API security. These vulnerabilities jeopardize the confidentiality, integrity, and availability of patient data, necessitating urgent and effective countermeasures. Moreover, we unveiled limitations in current security measures, particularly the deficiencies in documentation and monitoring protocols, which could impede healthcare organizations' ability to promptly detect and respond to security breaches.

Our research serves as a foundational resource for a diverse range of stakeholders, including healthcare providers, software developers, policymakers, and regulators. By disseminating this knowledge, we empower these stakeholders to make informed decisions aimed at enhancing healthcare API security and fortifying patient data privacy. Armed with these insights, they can implement essential tools and strategies to identify, mitigate, and prevent potential security breaches, thus fostering trust in healthcare services and ensuring the protection of invaluable patient information.

V. FUTURE SCOPE

While this research paper has effectively captured the current state of API security challenges in the Healthcare industry, there remain several avenues for further exploration and development. The following areas present opportunities for future research and improvements:

1. **Security Frameworks:** Future studies could focus on developing robust and standardized security frameworks specifically tailored for healthcare APIs. These frameworks should encompass best practices for authentication, encryption, and access control, facilitating seamless integration and consistent security implementations across different healthcare systems.

2. **Real-world Case Studies:** Conducting in-depth case studies of actual security breaches in healthcare APIs would provide valuable insights into the specific vulnerabilities and attack vectors prevalent in the industry. Analyzing real-world incidents can aid in understanding the root causes of security failures and guide the formulation of targeted security measures.

3. **Machine Learning and AI:** Exploring the potential of integrating machine learning and artificial intelligence into healthcare API security can significantly improve threat detection and response mechanisms. Developing intelligent systems capable of identifying anomalous behavior and adapting to emerging threats would strengthen the overall security posture of healthcare systems.

4. **Interoperability and Standardization:** As the healthcare industry continues to adopt digital transformation, ensuring interoperability and standardization of APIs becomes imperative. Future research can delve into creating interoperable API designs that prioritize security and facilitate seamless data exchange across different healthcare platforms.

5. **Regulatory Implications:** Investigating the impact of existing and potential future regulations on healthcare API security is essential. Understanding how compliance requirements influence security practices will enable organizations

TABLE II
LITERATURE REVIEW SUMMARY

Paper Citation	Main focus	Key-finding	Limitation
[1]	- Harnessing Healthcare APIs for Enhancing Interoperability: Ensuring API Security and Privacy	- Provides insights into API data integration and benefits for patients' medical records. - Identifies research questions to advance API security knowledge.	- Lacks comprehensive exploration of specific security measures. - Limited discussion on privacy and data exposure risks.
[2]	- Health Informatics on FHIR: How HL7's New API is Transforming Healthcare	- Comprehensive view of FHIR's potential impact in healthcare. - Practical relevance with use cases and case studies.	- Limited discussion on specific security challenges in healthcare.
[3]	- The Future of Blockchain Technology in Healthcare Internet of Things Security	-Explores benefits and shortcomings of Blockchain in healthcare. - Identifies future challenges for researchers.	- Lacks concrete case studies of Blockchain in healthcare.
[4]	- Malware Detection for Healthcare Data Security	- Effective approach for malware detection using ML techniques. - Emphasizes importance of protecting medical data.	- Does not explicitly discuss API security challenges in healthcare.
[5]	Early Prediction of Ransomware API Calls Behavior Based on GRU-TCN in Healthcare IoT	- Proactive approach to predict ransomware behavior in healthcare IoT. - Reduces likelihood of system infections.	- Lacks clear connection to API security challenges in healthcare.
[6]	- A Multi-View Attention-Based Deep Learning Framework for Malware Detection in Smart Healthcare Systems	- Novel deep learning framework for malware detection in healthcare systems. - Addresses malware detection effectively.	- Does not directly discuss API security challenges in healthcare.
[7]	- Blockchain for Healthcare Management Systems: A Survey on Interoperability and Security	- Comprehensive survey of architectural mechanisms for Blockchain-based HMS. - Offers valuable insights for improving healthcare interoperability and security.	- Needs more focus on API security challenges in healthcare.
[8]	- The Adoption of APIs for Digital Communications and the Implications for Cyber Security Vulnerabilities	- Emphasizes proactive security measures. - Compares different perspectives on API security.	- Could benefit from specific discussion on healthcare API security.
[9]	- SFTSDH: Applying Spring Security Framework With TSD-Based OAuth2 to Protect Microservice Architecture APIs	-Practical implementation of a hybrid security solution for healthcare APIs. -Complies with GDPR for data protection.	- Lacks specific discussion on API security challenges in healthcare.
[10]	- Applying Spring Security Framework with KeyCloak-Based OAuth2 to Protect Microservice Architecture APIs: A Case Study	- Integrated security solution for protecting healthcare APIs. - Successful implementation in a real-world scenario.	-Should address specific API security challenges in healthcare.
[13]	- Addresses IoT security challenges, particularly network intrusion detection systems (NIDS) in the IoT context.	- Categorizes IoT security threats, reviews existing NIDS tools and datasets, and analyzes NIDS proposals for IoT. Similarly, The growing use of IoT devices in healthcare underscores the importance of robust security measures, including API security, to protect sensitive healthcare data.	- Narrow focus on NIDS and machine learning techniques, potentially missing broader aspects of IoT security.
[14]	- Concentrates on Android security challenges, especially in the context of malware.	- Underscores the limitations of conventional security measures against Android malware. Likewise, Android's widespread use in healthcare highlights the crucial role of robust API security in healthcare apps and devices.	- May not delve deeply into technical details or offer specific remedies for Android security issues.
[15]	- Introduces "AndroSimilar," a technique for generating robust signatures to detect malicious Android applications.	- Proposes a signature-based method effective against code obfuscation and repackaging. Similarly, safeguarding healthcare APIs from malware variants is critical for protecting sensitive healthcare data.	- Potential limitations could involve the effectiveness of signature-based methods against highly sophisticated malware.
[16]	- Tackles the challenge of identifying portable executable (PE) malware by analyzing API call-grams.	-Proposes a semantic approach using API call-grams to detect previously unknown malicious code; which is extremely essential in healthcare as well. Protecting healthcare systems from PE malware threats underscores the need for advanced security measures, including API security, to ensure the integrity of healthcare APIs.	- Effectiveness of semantic analysis may vary depending on the diversity of malware samples.
[17]	- Evaluates the resilience of anti-malware techniques against code obfuscation in Android apps.	-Highlights the low resilience of anti-malware tools against code obfuscation. In healthcare, Android's prevalence in healthcare apps emphasizes the importance of robust API security in protecting sensitive healthcare data accessed through these apps.	- Effectiveness of anti-malware techniques can vary based on evolving obfuscation methods.
[18]	- Explores code obfuscation techniques in Android and their implications for security.	- Reviews code obfuscation methods and their relevance to healthcare app and API security. Likely, understanding code obfuscation is vital for safeguarding healthcare APIs that handle sensitive patient data.	- May not provide straightforward solutions to the challenges posed by code obfuscation.
[19]	- Presents "DroidAnalyst," an automated framework for app vetting and malware analysis on Android devices.	- The paper introduces a unified analysis model that combines static and dynamic analysis to enhance app code analysis, providing researchers and practitioners with a valuable tool for analyzing Android apps.. That may ultimately benefit the healthcare app security practitioners.	- Possible limitations could include the potential for false positives in malware detection and the need for continuous updates to adapt to the evolving Android ecosystem.
[20]	- Addresses the challenge of detecting malicious portable executable (PE) files through the analysis of API call-grams.	- The paper proposes a behavior model that focuses on extracting temporal snapshots of malware and benign executables using API call-grams, providing a means to detect dynamic and encrypted malware. That in turn may help protecting healthcare APIs and systems from malware threats.	- Behavioral analysis methods may exhibit an increased false alarm rate in healthcare settings, requiring careful tuning for healthcare-specific applications.

to proactively align their operations with regulatory standards [21].

6. User Education: Addressing the human factor in API security is crucial. Future research could explore effective methods of educating healthcare professionals and end-users about security best practices and potential risks, reducing the likelihood of security breaches caused by human errors.

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